COMPUTER PROGRAMMING

Assignment No. 1

Single C++ Statements

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Write a single C++ statement to accomplish each of the following:

Question i: Declare the variables voltage, current, and resistance to be of type double.

```
double voltage, current, resistance;   // Declare three double variables for voltage, current, and resistance
```

Question ii: Display the message "Welcome to Object Oriented Programming in C++!" on the screen.

```
cout << "Welcome to Object Oriented Programming in C++!";   // Output welcome message to the screen
```

Question iii: Read a value from the user and store it in an integer variable named age.

```
cin >> age;   // Read integer input from user into variable age
```

Question iv: If the variable temperature is greater than 100, print "Overheating detected!".

```
if (temperature > 100) cout << "Overheating detected!";   // Print warning if temperature exceeds 100
```

Question v: Compute the area of a circle with radius r and store it in area (π = 3.14159).

```
area = 3.14159 * r * r;   // Compute area using formula: area = π * r²
```

Question vi: Print the values of x, y, and z separated by a comma.

```
cout << x << "," << y << "," << z;   // Print x, y, z separated by commas
```

Question vii: Assign the sum of voltage and current to total.

```
total = voltage + current;   // Add voltage and current, store result in total
```

Question viii: Display "Error: Division by zero" only if denominator equals zero.

```
if (denominator == 0) cout << "Error: Division by zero";   // Display error only when denominator is zero
```

Question ix: Increment the variable count by 1 and then print its new value.

```
cout << ++count;   // Pre-increment count then print updated value
```

Question x: Read three floating-point numbers from the keyboard into a, b, and c.

```
cin >> a >> b >> c;   // Read three floating-point values into a, b, and c
```

Question xi: Total resistance in series (take values from user).

```
cin >> R1 >> R2 >> R3;   // Read three resistor values from user cout
<< "Total Resistance (Series) = " << (R1 + R2 + R3);   // Series:
R_total = R1 + R2 + R3
```

Question xii: Total resistance in parallel (take values from user).

```
cin >> R1 >> R2 >> R3;   // Read three resistor values from user
```

```
cout << "Total Resistance (Parallel) = " << 1 / ((1/R1) + (1/R2) + (1/R3));
// Parallel: 1/R_total = 1/R1 + 1/R2 + 1/R3
```

Question xiii: Voltage across R2 using Voltage Division Rule. (R1 = 10Ω, R2 = 20Ω, V = 12V)

```
cout << "Voltage across R2 = " << 12 * (20.0 / (10 + 20));   // Voltage
Division: V_R2 = V × (R2 / (R1 + R2))
```

Question xiv: Current through R2 using Current Division Rule. (R1 = 10Ω, R2 = 20Ω, V = 12V)

```
cout << "Current through R2 = " << (12.0 / (10 + 20)) * (10.0 / (10 + 20));
// Current Division: I_R2 = I_total × (R1 / (R1 + R2))
```

Question xv: Calculate Semester GPA.

```
cin >> credit >> grade;   // Read credit hours and grade points for each
subject
totalCredits += credit;  totalPoints += credit * grade;   // Accumulate
weighted grade points and credits
cout << "Semester GPA = " << (totalPoints / totalCredits);   // GPA = total
grade points / total credit hours
```